

Social Engineering & Security Awareness

Learn :

Social Engineering : It is the art of manipulating people to reveal their confidential information such as passwords, bank details or personal data.

→ Instead of hacking computers, they use human psychology by exploiting their trust, fear or urgency.

Types of Social Engineering :-

1. Phishing : Attackers send fraudulent emails, messages or website to trick users into revealing sensitive information such as login credentials, banking details etc

2. Pretexting : Attackers create a false identity or scenario to gain trust or extract information.

3. Baiting : Attackers lure victims by offering something appealing (free software, movies, jobs) that contains malware.

4. Tailgating (Piggybacking) : Attackers physically follow someone into a restricted area without proper authorization.

Task :

Research and summarize one real-world social engineering attack

Twitter Bitcoin Scam 2020 : -

Attack Type : Phishing & Impersonation(Whaling)

In 2020, hackers gained access to their Twitter's internal tools by targeting employees through a phishing attack.

They then hijacked the profiles of high profile users like Elon Musk, Barack Obama, Bill Gates, Apple and Kanye West.

How did the attack work?

1. Hackers send fake login pages to twitter's employees, tricking them into their login credentials.

2. They then reset password and bypassed 2FA(Two-Factor Authentication).

3. Using internal tools, they took over verified accounts and twitted a Bitcoin scam message.

"I am giving back to the community! Send Bitcoin to this address, and I will send back double."

4. Thousands of people fell for it, sending over \$118,000 before Twitter shut it down.

Write a short article/post explaining the importance of cybersecurity awareness.

In today's digital age, cyber threats like phishing, malware, and data breaches are growing rapidly. Cybersecurity awareness is crucial to protect personal and financial information from hackers.

Simple practices like using strong passwords, enabling multi-factor authentication (MFA), avoiding suspicious links, and keeping software updated can prevent major security breaches. Many cyber attacks succeed due to human error, making awareness the first line of defense.

By staying informed and cautious online, individuals and businesses can safeguard sensitive data and reduce cyber risks. Stay alert, stay secure!

Learn :

How phishing attack works :-

Hackers trick people into revealing sensitive information by pretending to be a trusted entity

Identifying phishing emails :-

- x Suspicious sender
- x Urgent or Threatening Language
- x Unexpected Attachments or Links
- x Poor Grammar or Spelling
- x Request for personal information

Email security best practices (SPF, DKIM, DMARC).

1. SPF (Sender Policy Framework) :-

- x What it does: Ensures only authorized mail servers can send emails on behalf of a domain.
- x How it helps: If an attacker tries sending an email from hacker@example.com using a different mail server, the recipient's system rejects or marks it as spam.

2. DKIM (Domain Keys Identified Mail) :-

- x What it does: Adds a digital signature to emails, proving they weren't modified during transmission.
- x How it helps: If an attacker modifies the email, the digital signature fails verification, marking the email as suspicious.

3. DMARK (Domain Based Message Authentication Reporting & Conformance) :

- x What it does: Tells email servers what to do with unauthorized emails (reject, quarantine, or monitor).
- x How it helps: If SPF or DKIM fails, the email is rejected instead of landing in the inbox.

Task :

Find and analyze a real phishing email (safely) or use phishing test websites

Use Phishing test website (Phish Tank)

Phish Tank

→ select any phishing website from the provided

PhishTank is operated by [Cisco Talos Intelligence Group](#)

[Home](#) [Add A Phish](#) [Verify A Phish](#) [Phish Search](#) [Stats](#) [FAQ](#) [Developers](#) [Mailing Lists](#) [My Account](#)

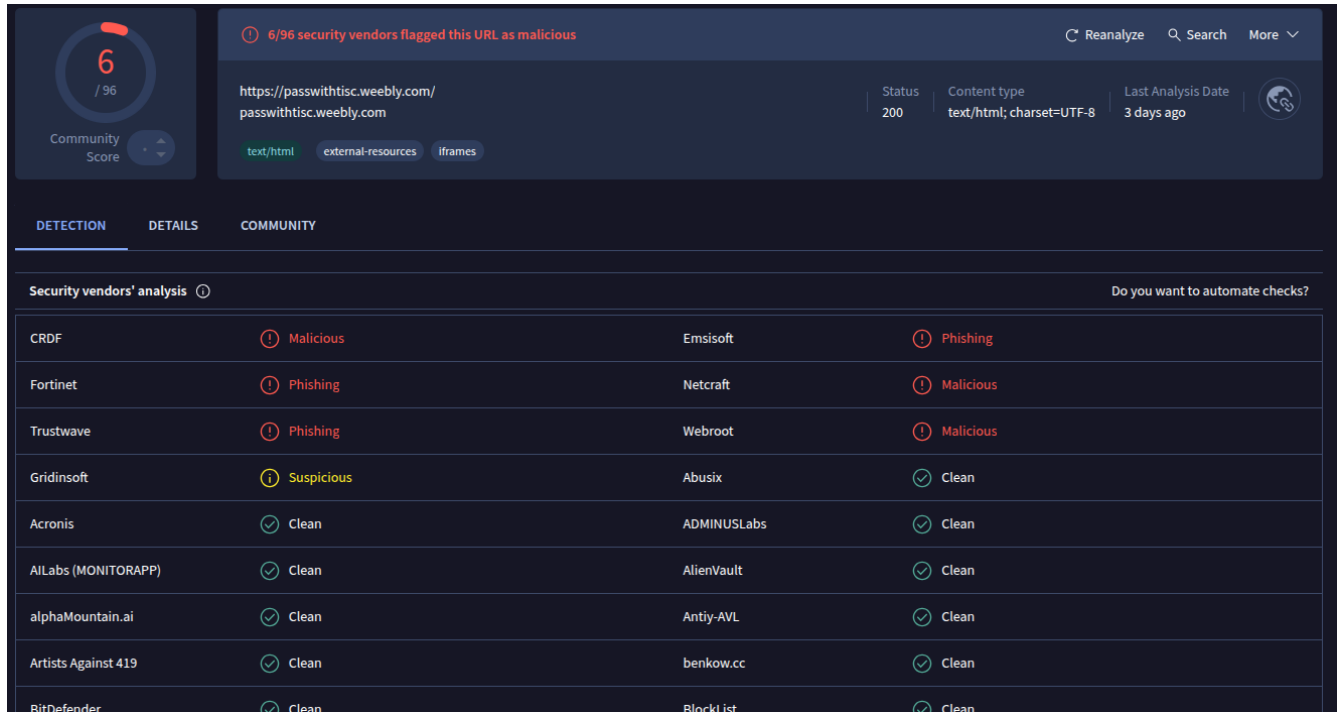
Verify A Phish

Showing unverified and online submissions
[See all submissions in the phish archive](#)

ID	Phish URL	Submitted	Valid?	Online?
9006362	https://casplendure.weebly.com/ added on Mar 7th 2025 4:09 PM	by buaava	Unknown	ONLINE
9006361	https://my-office-bit.weebly.com/ added on Mar 7th 2025 4:09 PM	by buaava	Unknown	ONLINE
9006360	https://mimemal.weebly.com/ added on Mar 7th 2025 4:09 PM	by buaava	Unknown	ONLINE
9006359	https://fopper-wku.weebly.com/ added on Mar 7th 2025 4:09 PM	by buaava	Unknown	ONLINE
9006357	https://valdabonrequired.weebly.com/ added on Mar 7th 2025 4:09 PM	by buaava	Unknown	ONLINE
9006356	https://micosdm4.godaddy.com/ added on Mar 7th 2025 4:09 PM	by buaava	Unknown	ONLINE
9006355	https://eltyu-107163.weebly.com/ added on Mar 7th 2025 4:07 PM	by buaava	Unknown	ONLINE
9006353	https://businessconfymagmail.weebly.com/ added on Mar 7th 2025 4:07 PM	by buaava	Unknown	ONLINE
9006352	https://mimdmnln.weebly.com/ added on Mar 7th 2025 4:07 PM	by buaava	Unknown	ONLINE
9006349	https://fellschueba.weebly.com/ added on Mar 7th 2025 4:07 PM	by buaava	Unknown	ONLINE
9006348	https://pagesatt.weebly.com/ added on Mar 7th 2025 4:07 PM	by buaava	Unknown	ONLINE
9006347	https://glbaw.weebly.com/ added on Mar 7th 2025 4:06 PM	by buaava	Unknown	ONLINE
9006346	https://mh011-service001.weebly.com/ added on Mar 7th 2025 4:06 PM	by buaava	Unknown	ONLINE
9006345	https://passentilac.weebly.com/ added on Mar 7th 2025 4:06 PM	by buaava	Unknown	ONLINE
9006344	https://mp0nrbentheat.weebly.com/ added on Mar 7th 2025 4:06 PM	by buaava	Unknown	ONLINE
9006343	https://28369417245n5connect.weebly.com/ added on Mar 7th 2025 4:05 PM	by buaava	Unknown	ONLINE
9006342	https://mapchatunlocked1.weebly.com/ added on Mar 7th 2025 4:05 PM	by buaava	Unknown	ONLINE

Report of <https://passwithtisc.weebly.com/>

1. analyze it on Virus total



The screenshot shows the VirusTotal interface for the URL <https://passwithtisc.weebly.com/>. The top section displays a "Community Score" of 6/96, indicating a high risk. A warning message states "6/96 security vendors flagged this URL as malicious". The URL is listed as "https://passwithtisc.weebly.com/passwithtisc.weebly.com" with a status of 200 and content type of "text/html; charset=UTF-8". The last analysis was performed 3 days ago. Below this, the "DETECTION" tab is active, showing a table of security vendors' analyses. The table lists 19 vendors, with 6 flagged as malicious, phishing, or suspicious, and 13 marked as clean.

Security vendors' analysis		Do you want to automate checks?	
CRDF	Malicious	Emsisoft	Phishing
Fortinet	Phishing	Netcraft	Malicious
Trustwave	Phishing	Webroot	Malicious
Gridinsoft	Suspicious	Abusix	Clean
Acronis	Clean	ADMINUSLabs	Clean
ALabs (MONITORAPP)	Clean	AlienVault	Clean
alphaMountain.ai	Clean	Antiy-AVL	Clean
Artists Against 419	Clean	benkow.cc	Clean
BitDefender	Clean	Blocklist	Clean

6 out of 96 security vendors flagged this URL as:

Malicious (CRDF, Netcraft, Webroot)

Phishing (Fortinet, Trustwave, Emsisoft)

Suspicious (Gridinsoft)

90 vendors marked it as clean, but even a few phishing/malicious detections are a major red flag!

2. on Whois.com

weebly.com		Updated 14 hours ago ↻
Domain Information		
Domain:	weebly.com	
Registered On:	2006-03-29	
Expires On:	2025-03-28	
Updated On:	2024-12-10	
Status:	client delete prohibited client transfer prohibited client update prohibited	
Name Servers:	ns-123.awsdns-15.com ns-1500.awsdns-59.org ns-1797.awsdns-32.co.uk ns-646.awsdns-16.net	
Registrar Information		
Registrar:	SafeNames Ltd.	
IANA ID:	447	
Abuse Email:	abuse@safenames.net	
Abuse Phone:	+44.1908200022	

Phishing Risk?

- ✗ If someone reported a phishing attempt related to Weebly, it could be due to a spoofed or lookalike domain (e.g., **weebIy.com** with a capital "I" instead of "l").
- ✗ The presence of similar-looking domains for sale (weeblynews.com, weeblym.com, etc.) suggests that phishing actors might register fake domains to impersonate Weebly.

~ Task 5 Week 5

~ March 7, 2025